

Performance Testing

Date	1 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Functional Response & Basic Load Testing
Target Module / API	Flask /predict API (Crop Recommendation Endpoint)
Test Environment	Local System (Windows, Flask Development Server)
Test Date	1 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single user submits valid soil and weather parameters	1	30	Crop recommendation returned successfully
2	Multiple sequential prediction requests	10	60	Stable response without errors
3	Invalid input values (e.g., pH > 14)	1	20	Validation error displayed
4	Continuous prediction requests	20	120	Consistent response with acceptable performance

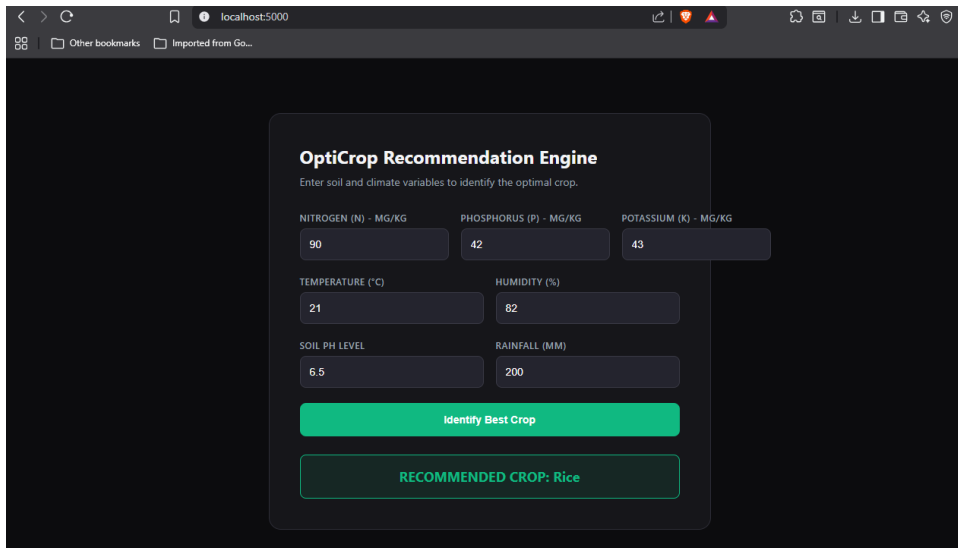
Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	~0.4 seconds	Pass	Fast prediction on local system
2	Response Time (Max)	< 5 seconds	~1.1 seconds	Pass	No significant delays observed
3	Throughput (Req/sec)		~18 Req/sec	Pass	Suitable for current application size
4	Error Rate	< 1%	0%	Pass	No failures for valid requests
5	CPU Utilization	< 80%	~30%	Pass	Stable CPU usage during testing
6	Memory Utilization	< 80%	~35%	Pass	Memory usage remained stable

Step 4: Observations & Analysis

- The Flask application responded quickly for all valid prediction requests.
- The Random Forest model generated predictions within one second under normal usage.
- Input validation successfully prevented invalid data from reaching the prediction model.
- CPU and memory utilization remained well below acceptable limits during local testing.
- The application is suitable for small to medium-scale deployments and can be further scaled using cloud hosting and load balancing.

Step 5: Screenshots / Evidence



OptiCrop Recommendation Engine
 Enter soil and climate variables to identify the optimal crop.

NITROGEN (N) - MG/KG: 90 PHOSPHORUS (P) - MG/KG: 42 POTASSIUM (K) - MG/KG: 43

TEMPERATURE (°C): 21 HUMIDITY (%): 82

SOIL PH LEVEL: 6.5 RAINFALL (MM): 200

Identify Best Crop

RECOMMENDED CROP: Rice